

Respectful collaboration starts here



ALL ARE
WELCOME

TODO
BIENVENIDOS

みなさん
歓迎します。

www.lfdecentralizedtrust.org/code-of-conduct

Hiero Heka: A Ready-To-Use Decentralized Identity Solution

LFDT Meetup

Presented by

Alexander Shenshin – DSR Corporation

05/21/2026

Agenda

- About LFDT Hiero project
- Decentralized Identity, use cases and adoption
- Heka Identity Platform
- Integration with Hiero Ledger
- Demo
- Roadmap and long-term vision
- Q&A

LF DECENTRALIZED TRUST



What is Hiero?

Hiero is a [Linux Foundation Decentralized Trust](#) project, the first Layer 1 platform to be openly governed by a trusted independent foundation.

It is the open-source, vendor-neutral distributed ledger technology that is used to build the [Hedera](#) public ledger.

Why Linux Foundation?

- Trusted Brand
- Established Foundation of related projects including Besu
- Pathways to Cross-Chain Interoperability
- Expanding Hyperledger Foundation to Decentralized Trust



Open Source Distributed Ledger Technology

LF DECENTRALIZED TRUST

The Layers of the Hiero / Hedera Ecosystem



LEVEL 3: APPS ON HEDERA



LEVEL 2: HEDERA



LEVEL 1: HIERO

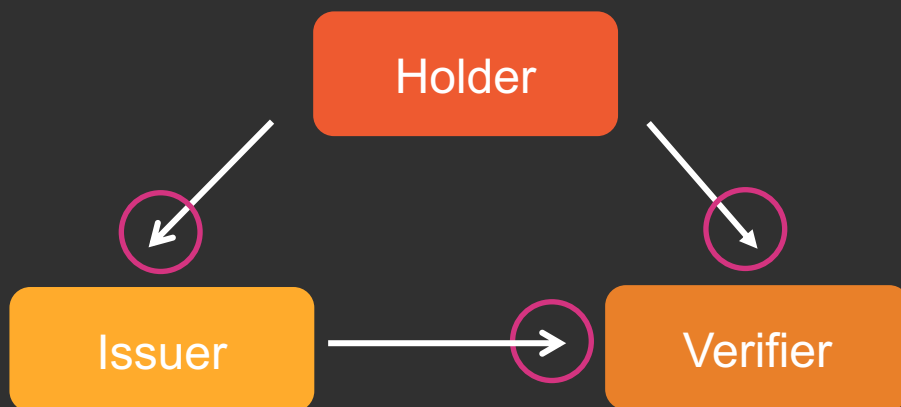


Decentralized Identity



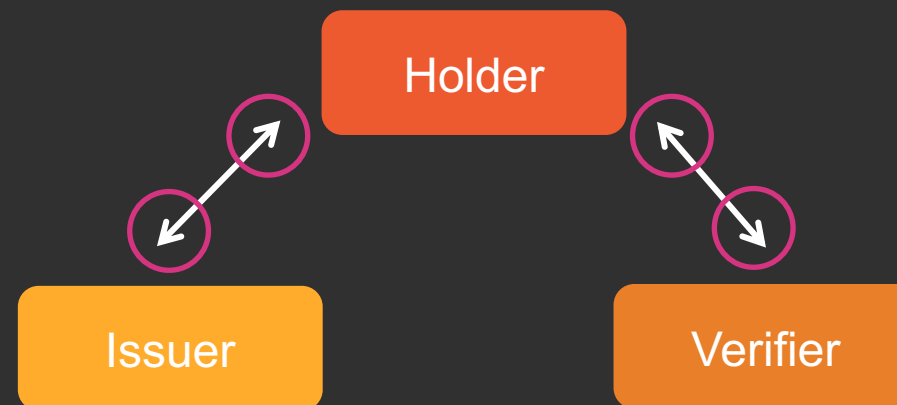
- Focus on **Privacy**
- **User-controlled** identity that doesn't depend on any single vendor
- Multiple **standards** and frameworks that implement the principles
- Adoption by the **Governments** (EU, USA, Japan, Canada, Bhutan, etc.) and **BigTech** (Google, Apple, Microsoft, etc.)

Federated vs Decentralized Identity



Federated Identity (OAuth, OpenID)

Authorization-based credential exchange where the holder authorizes a verifier (client) to access information on his/her behalf.



Decentralized Identity (OID4VC, etc.)

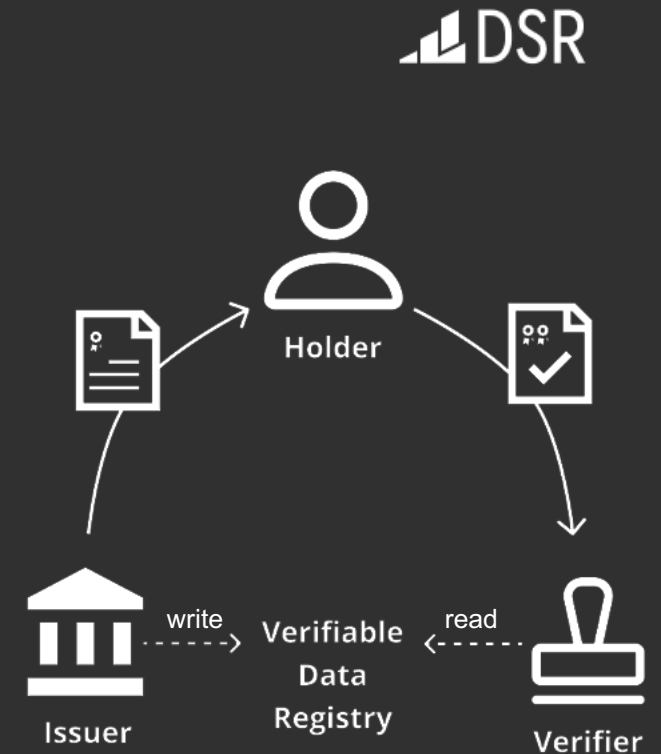
Self-sovereign credential exchange where the holder can autonomously control the exchange of credentials with any verifier he/she wants.

Decentralized Identity Example

1. Alice installs a certified Mobile Wallet application on her phone
2. Government issues a digital ID (passport or driver license) for Alice. The digital ID is stored in Alice's Mobile Wallet.
3. Alice is getting on an airplane, and needs to present her ID. She presents her digital ID from the Mobile Wallet.

As we can see Alice is in control of her information and decides what parts of that information other entities can see.

Unlike a traditional system, the airport never contacted the government — it just verified Alice's signed credential offline.

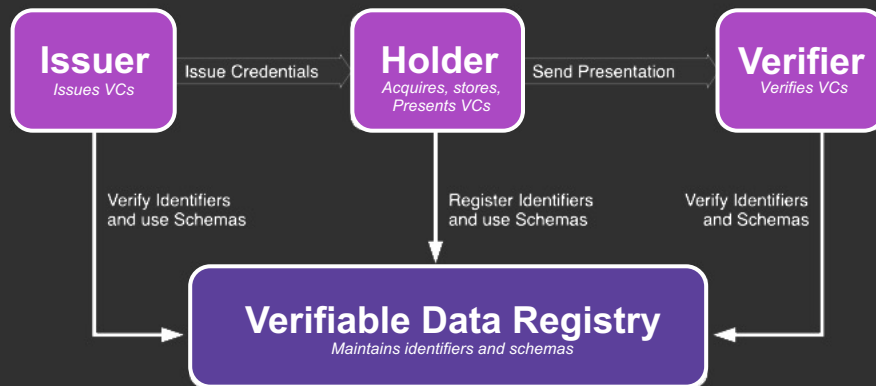


- Alice - Holder
- Government - Issuer
- Airport - Verifier

Decentralized Identity Technical Concepts

1. Verifiable Credentials (VC)

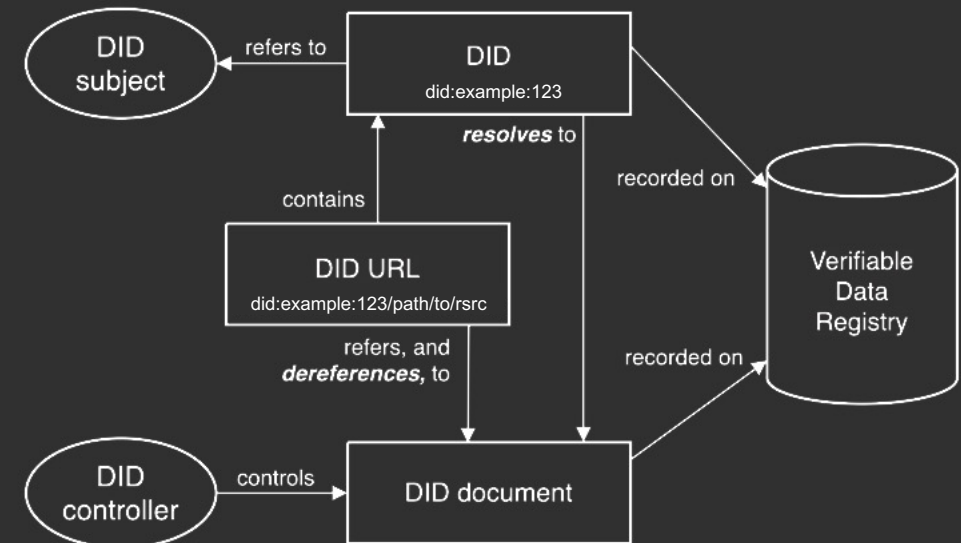
Verifiable Credentials Data Model v1.1,
W3C Recommendation 2022



From <https://www.w3.org/TR/vc-data-model/>

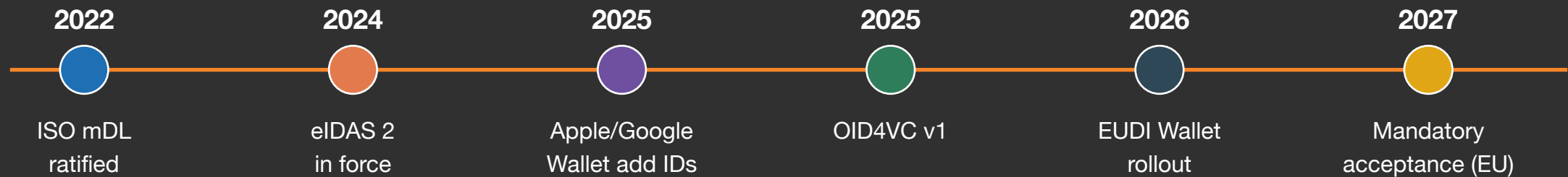
2. Decentralized Identifiers (DID)

Scheme
did : example : 123456789abcdefghijk
DID Method DID Method Specific String



From <https://www.w3.org/TR/did-core/>

Decentralized Identity Adoption timeline



Standards

W3C VC 2.0 · DIDs 1.0
OID4VC v1.0 · IETF SD-JWT VC · ISO mDoc

Wallets

Apple · Google · EUDI
Heka · OWF Bifold

Issuers

Primarily governments and banks
13+ US states ship mDL
Every EU member state by 2026

Decentralized Identity Use Cases



Fintech

- Reusable **KYC/AML** onboarding
- **Portable identity** for ecosystems
- Prevents centralized **data breaches**
- Prevents **identity fraud** & impersonation

AI

- **Verifiable interaction** by AI agents
- Cryptographic content **provenance**
- Verifiable **audit trails** for AI
- Privacy-preserving AI **training** data

Healthcare

- **Portable identity** for patients
- Prevents **PHI** exposure or tampering
- Instant **medical license** verification

IoT, Energy and Automotive

- Verifiable **access control**
- Secure **identity** for devices & assets
- **Ownership** & lifecycle management

Public Sector

- Digital **citizen identity**
- **Fraud** reduction
- Seamless **cross-border** ID verification

Telecom and Entertainment

- Prevents **voice phishing**
- Digital **copyrights** & ownership
- Secure **age verification**
- Accelerated **user onboarding**

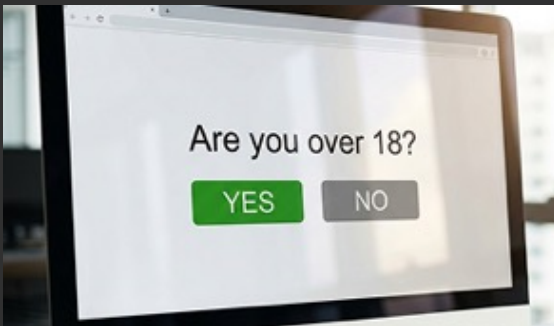
Hiero Heka Focus: Core and Emerging Use Cases



Decentralized KYC/KYB:
Enabling Cross-Institution Identity Portability
#fintech #identity #web3



Empowering the Agentic Economy:
Granular API Monetization via Stablecoins and Verifiable Credentials
#fintech #identity #web3 #ai #stablecoins



Privacy-Preserving Age Verification
#public-sector #telecom #identity #zero-knowledge



Agent to Agent Step-Up Authorization:
Decentralized Trust Model for A2A interactions via Verifiable Credentials
#identity #ai #a2a

Decentralized Identity in LFDT Hiero — Why Heka?



1

Foundational use case

Decentralized Identity is one of DLT's foundational use cases, increasingly driven by global interoperability standards.

2

Hiero's identity journey

Years of community work — starting with the Hedera DID Method, then broader support in OWF ACA-Py and Credo.

3

The remaining gap

Ready-to-use, end-user applications are still scarce — crucial for broader DLT adoption.

Heka Identity Platform — ready-to-use, reference-grade.

Heka Identity Platform



A ready-to-use decentralized identity solution for the Hiero ecosystem — published under LFDT Hiero by DSR Corporation, based on DSR SSI Toolkit and leveraging OWF Credo and Bifold.

Batteries-included

Issuer · Holder · Verifier components and UI delivered out of the box.

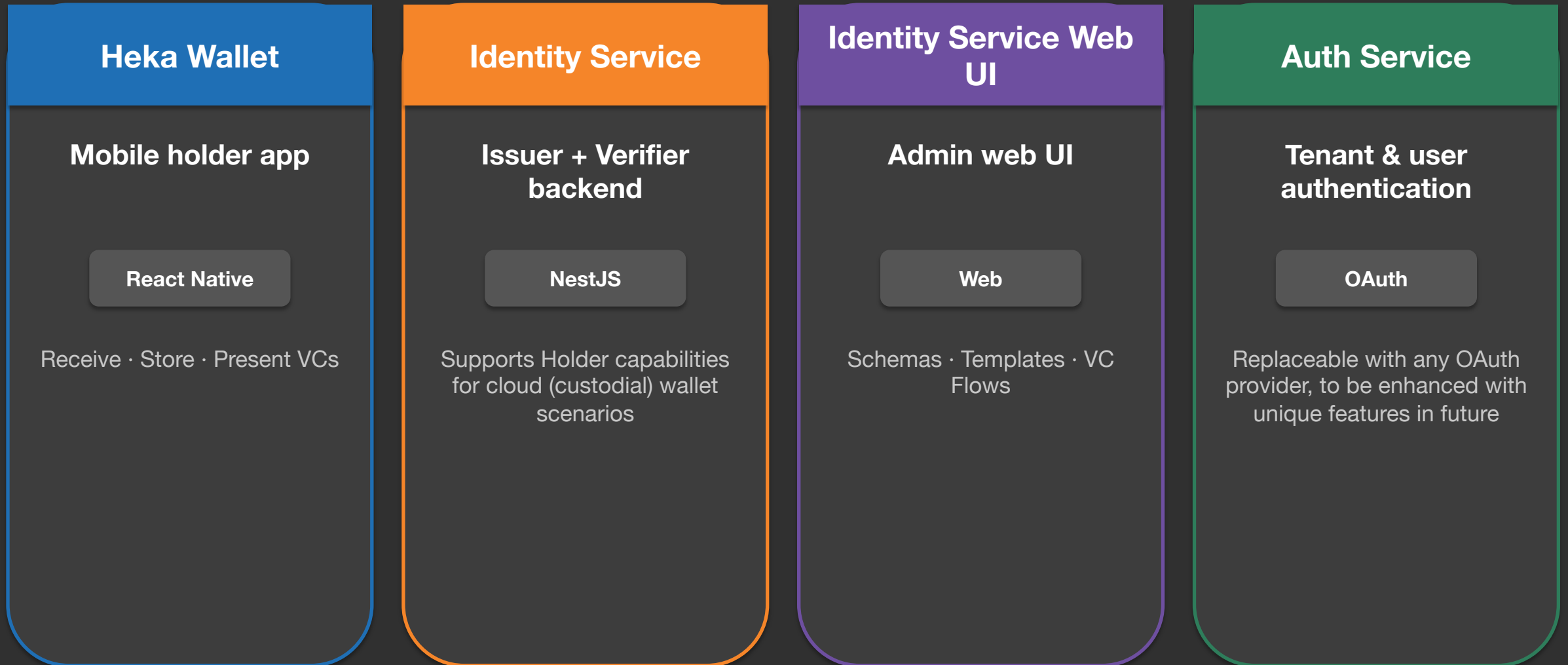
Modular & white-label

Extensible API-first architecture designed for customization and re-branding.

Advanced features

Role model, wallet backup with passkeys, crypto wallet support, and more.

Platform Components



Supported Standards



Credential Exchange Protocols

- OpenID for Verifiable Credentials (OID4VC)
- DIDComm

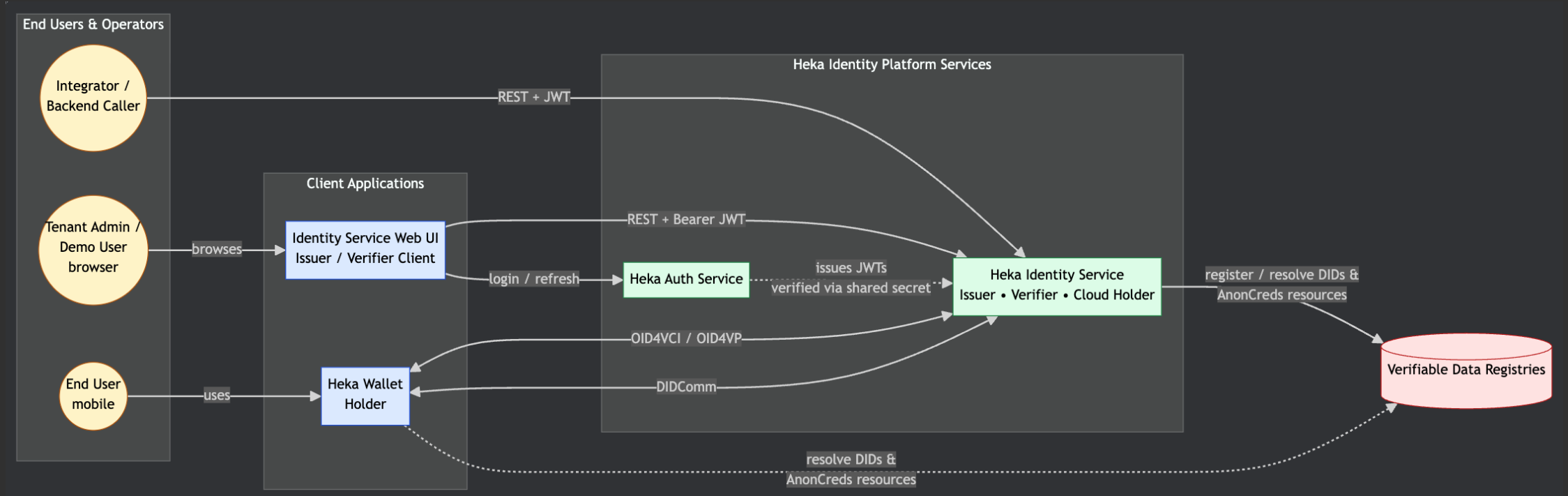
Credential Formats

- W3C VC-JWT
- W3C VC-JWT JSON-LD
- W3C VC with Linked Data Proofs
- IETF SD-JWT VC
- ISO mDoc (mDL)
- Hyperledger AnonCreds

DID Methods

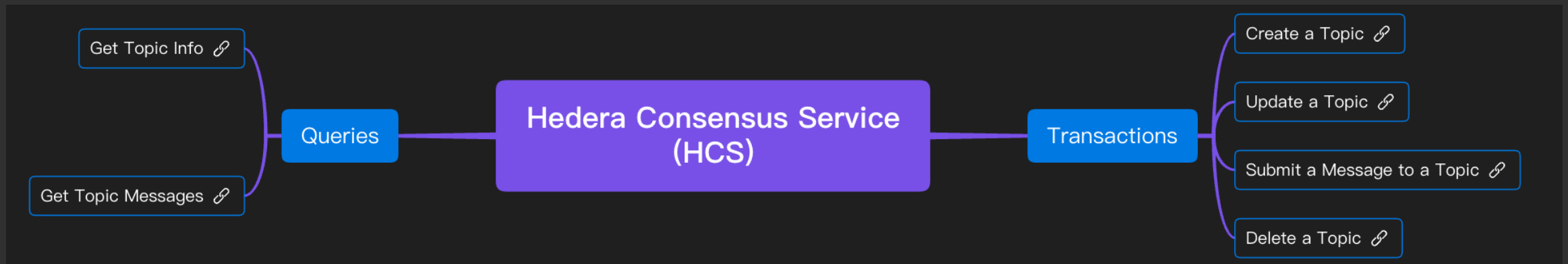
- did:key
- did:peer
- did:jwk
- did:web
- did:indy (Hyperledger Indy)
- did:hedera (Hiero / Hedera)
- did:indybesu (Indy Besu)

High-level Platform Architecture



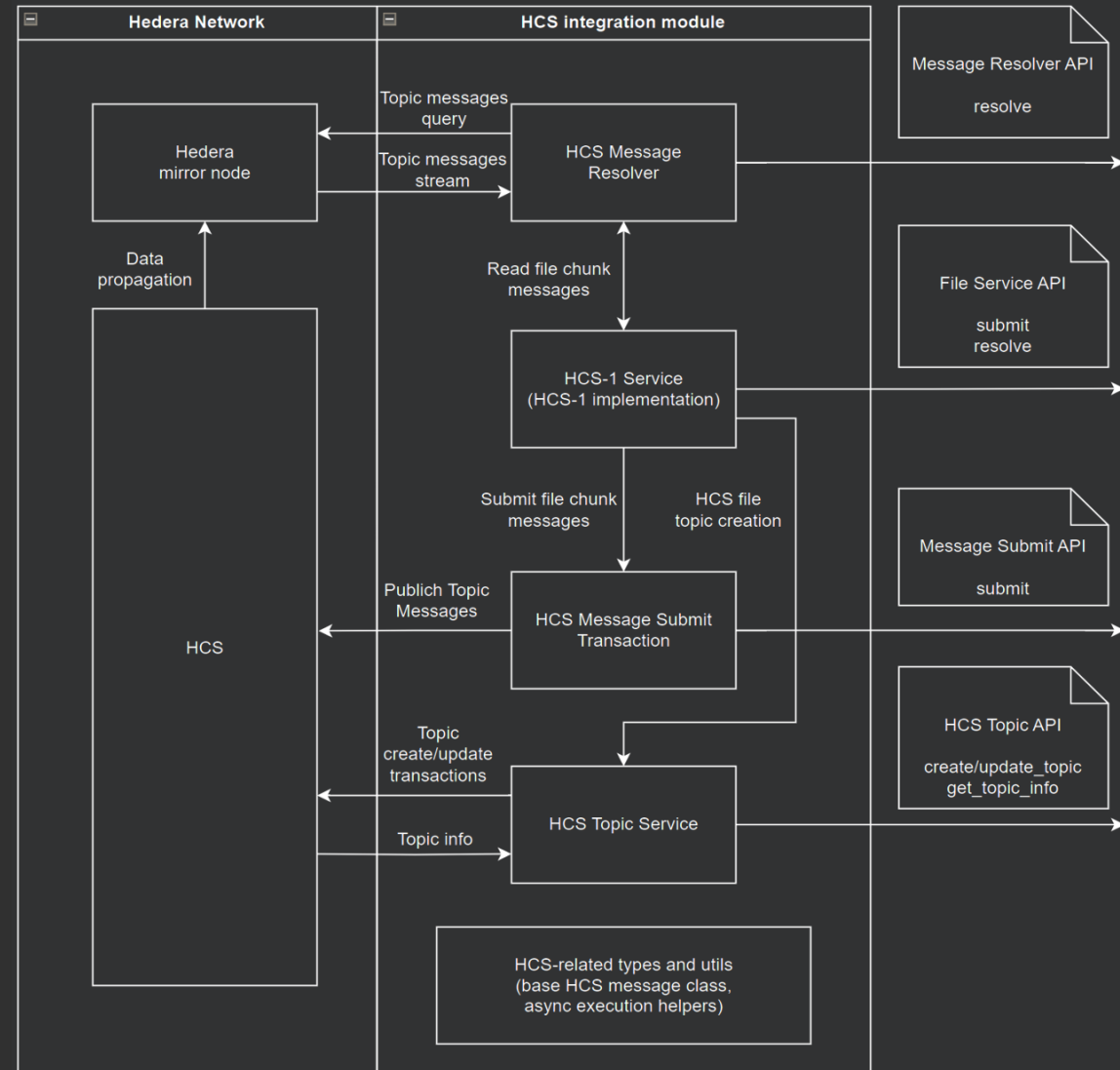
Hiero / Hedera Consensus Service

- User can create topics and then publish messages to the topic
- Messages can contain any payload up to 1024 bytes
- High throughput, highly scalable, and low cost



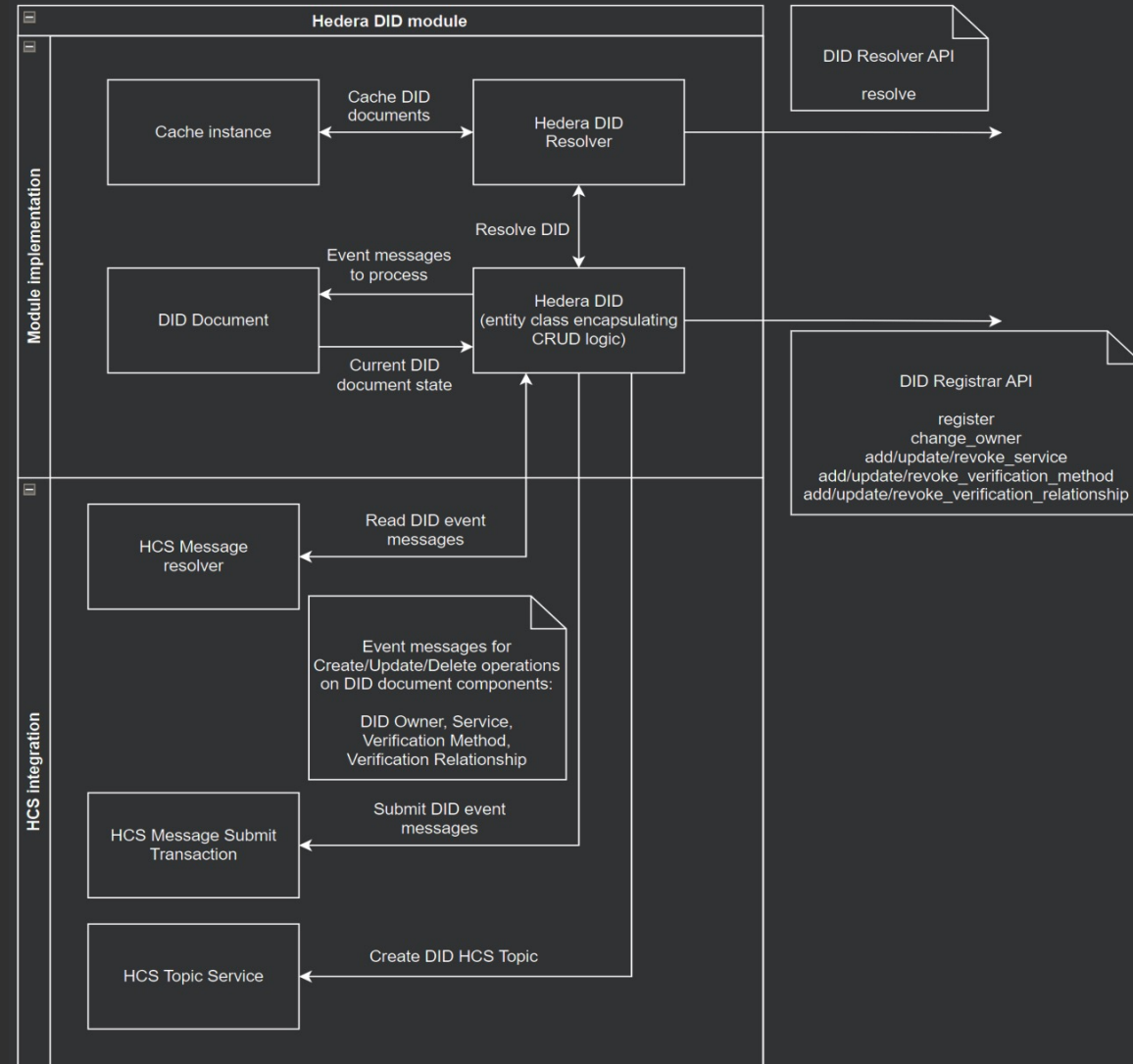
Integration with Hiero Ledger

- Under-the-hood integration with Hiero-based network (Hedera)
- Main components/layers:
 - Hedera network
 - Hiero Consensus Service (HCS) integration module
 - HCS integration API (used by other modules)



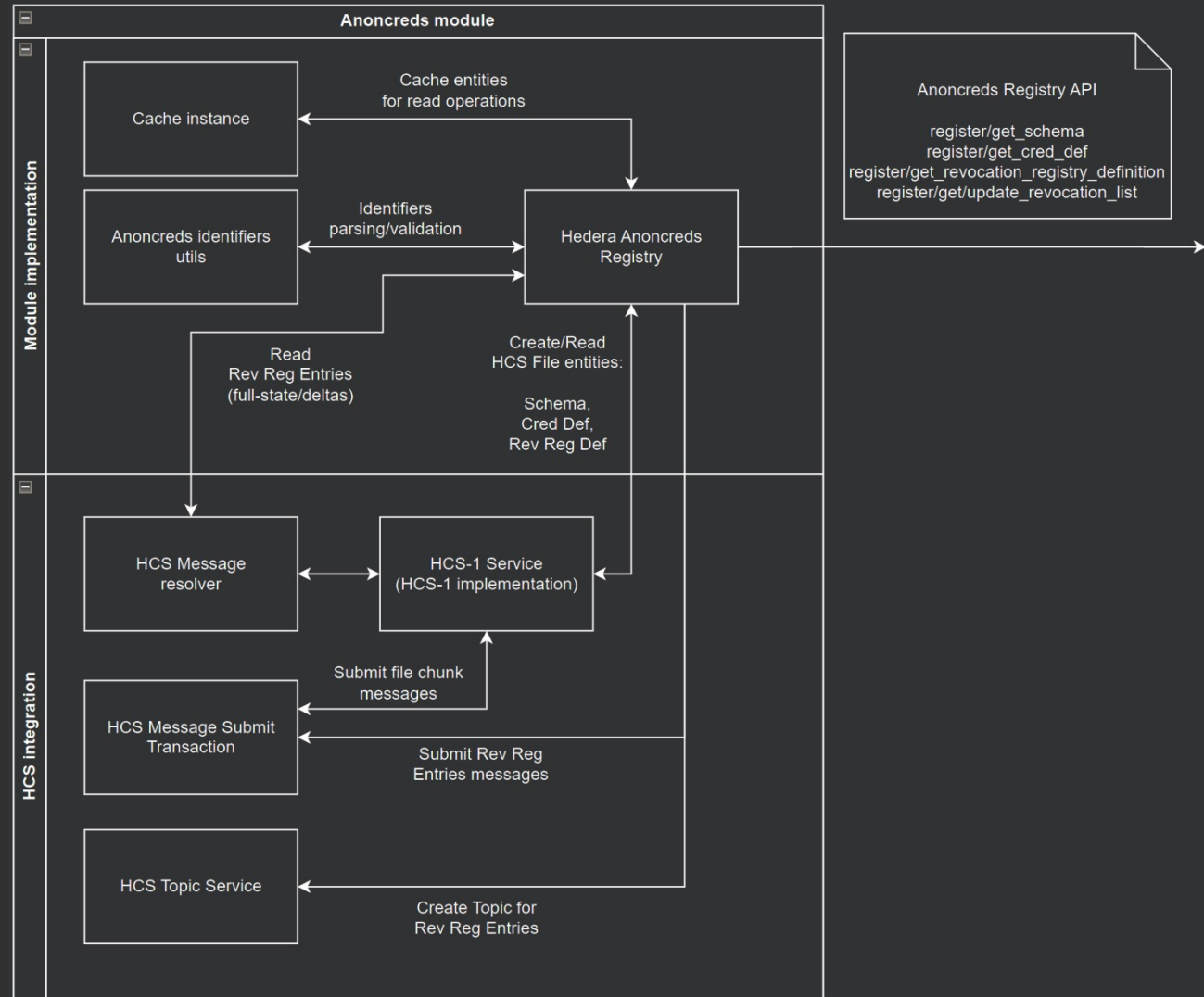
Integration with Hiero Ledger

- Hedera DID management and integration with HCS
- Main components/layers:
 - Hiero Consensus Service (HCS) integration module
 - Hedera DID module
 - Hedera DID resolver and registrar APIs



Integration with Hiero Ledger

- AnonCreds resources management and integration with HCS
- Main components/layers:
 - Hiero Consensus Service (HCS) integration module
 - AnonCreds module
 - AnonCreds registry API



- Demos for core and emerging identity use cases:
 - Decentralized KYC: Enabling Cross-Institution Identity Portability
 - Privacy-Preserving Age Verification
 - Agent to Agent Step-Up Authorization: Decentralized Trust Model for A2A interactions via Verifiable Credentials
 - Empowering the Agentic Economy: Granular API Monetization via Stablecoins and Verifiable Credentials
- All 4 demos are built with Heka Identity Platform and leverage its components (fully or partially depending on a use case)

Demo: Decentralized KYC



Cross-Institution Identity Portability — a user completes KYC once, receives a Verifiable Credential, and reuses it across institutions without repeating onboarding.



Reusable KYC via Decentralized Identity:

See how a user completes KYC once, receives a verifiable credential, and instantly verifies with another bank without repeating the process.

Demo: Privacy-Preserving Age Verification



Secure and privacy-preserving age verification based on Mobile Driver License (mDL) credential presentation. The demo leverages Heka Identity Platform and showcases support for MSO mDoc credential format with selective disclosure of claims.



**Privacy-first age verification
using a mobile driver's license**

based on Heka Identity Platform

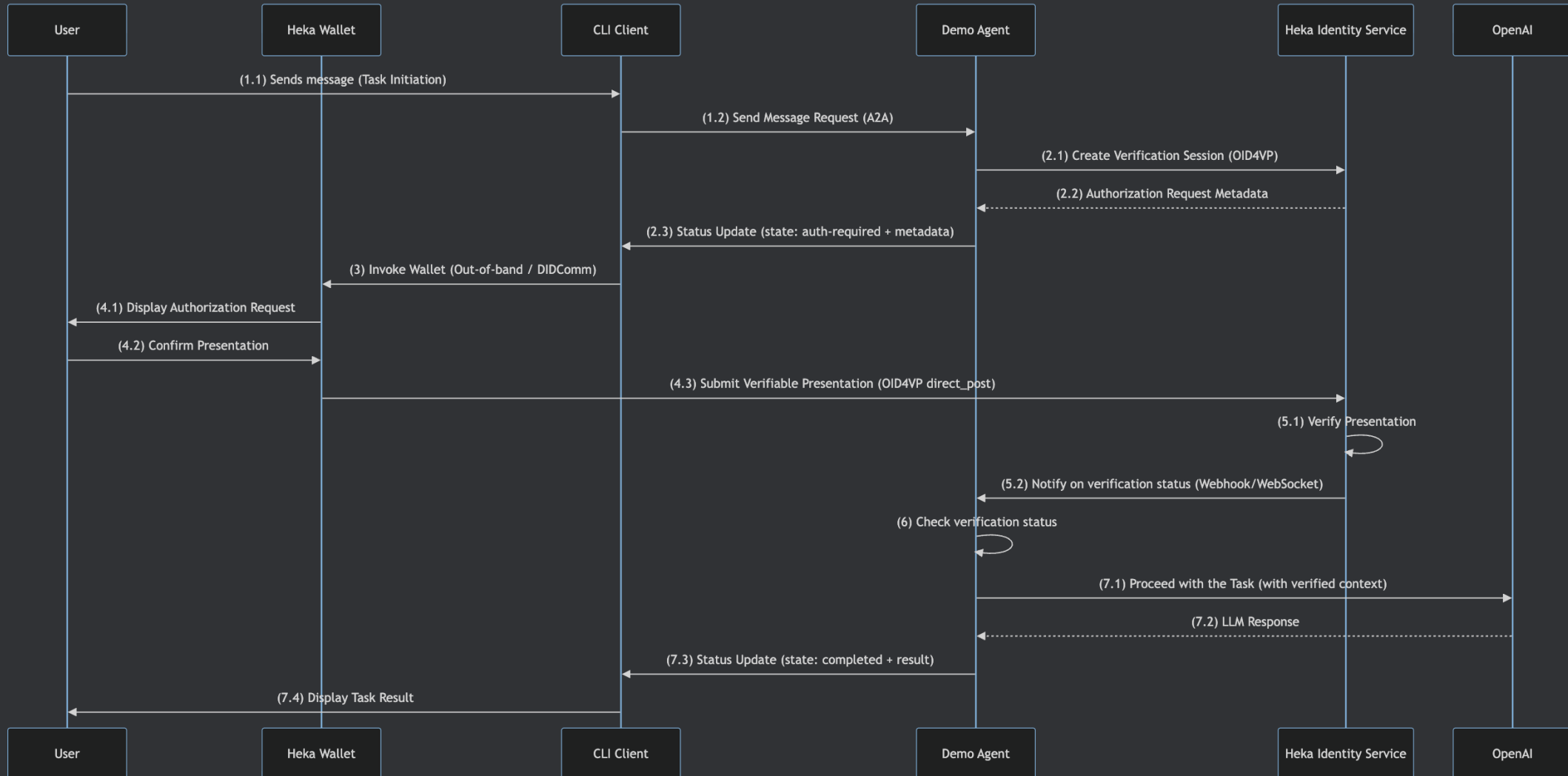
Demo: A2A Step-Up Authorization



Decentralized Trust Model for A2A — VC-based authorization for interactions between AI Agents. The demo implements the OID4VP In-Task Authorization extension for A2A, enabling Verifiable Credential presentations as task authorization option.



A2A Step-Up Authorization – How it works



Demo: Empowering the Agentic Economy



Granular API monetization — AI agents paying for digital services via stablecoins. A user assigns a task and budget to an AI agent, issuing a Verifiable Credential to authorize agent action in a fine-grained way. The agent interacts with external services via APIs and pays per request using stablecoins.



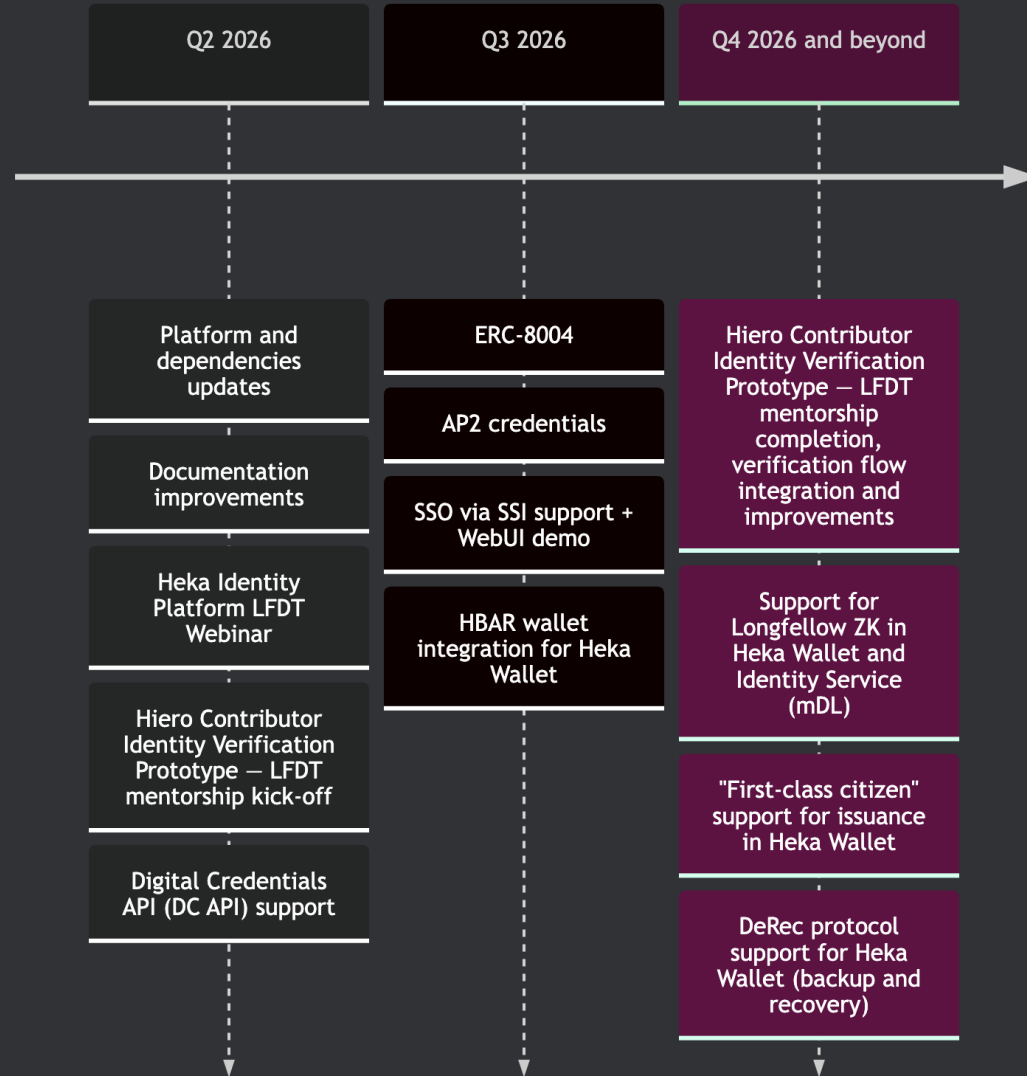
AI Agent Payments via Stablecoins

x402 Monetization Wrapper for APIs

Hiero Heka Roadmap

High-level scope:

- Core maintenance, platform updates and community support
- Foundational support for modern AI & agentic economy use cases and protocols
- Continued prototyping and integration of identity solutions in Hiero ecosystem
- Advanced ZKP support (Longfellow ZK)



Long-term vision



- Focus on emerging synergy between **Agentic AI** and **Decentralized Identity**
 - Support for relevant protocols and standards (**A2A**, **AP2**, **Verifiable Intent**, **x402**) as **core features of the platform**
 - Use cases: **Agent Economy**, Decentralized Trust for **multi-agent ecosystems** and agent to agent interactions
- Integration with **broader Hiero ecosystem** is a crucial part of the roadmap
 - Gradual evolution of identity from relevant supporting capability to **one of core platform layers** for projects / products built on Hiero
 - Providing "**identity rail**" options for relevant DLT use cases (including **KYC / KYB**, **Asset Tokenization**, and **AI**)
 - Current focus: **Hiero Contributor Verification prototype** and its evolution (LFDT mentorship program), **Hiero-specific identity features** and exploring further cross-project integrations

Join the community



LFDT Discord

Hyperledger / LFDT community-wide server



discord.com/invite/hyperledger

Hiero & Heka

Hiero channels + #heka-identity-platform



discord.gg/VPa3k4y5KM

Hiero Identity Community Meetings

Register for Hiero Identity Community calls



zoom-lfx.platform.linuxfoundation.org

Technical resources

- Hiero GitHub organization: <https://github.com/hiero-ledger>
- Repositories
 - Heka Identity Platform: <https://github.com/hiero-ledger/heka-identity-platform>
 - Hiero DID SDK JS: <https://github.com/hiero-ledger/hiero-did-sdk-js>
 - For architecture/implementation details, please see design docs located in the repos
- Demos
 - A2A OID4VP In-Task Authorization
 - Recording: <https://www.youtube.com/watch?v=3JgFZBGXXI>
 - Source code: <https://github.com/hiero-ledger/heka-identity-platform/tree/main/demo/a2a-oid4vp>
 - AI Agent Payments via Stablecoins: <https://www.youtube.com/watch?v=ef6NAwDavb0>

Hiero Heka Repository



Scan to explore Hiero
Heka repo

Technical resources



- Demos
 - Decentralized KYC: <https://youtu.be/giMWdOU62ro>
 - Privacy-preserving Age Verification: <https://youtu.be/2hEKOoza4uQ>
- Specifications
 - Hedera DID Method spec: <https://github.com/hashgraph/did-method/blob/master/hedera-did-method-specification.md>
 - Hiero AnonCreds Method spec: <https://hiero-ledger.github.io/identity-collaboration-hub/hiero-anoncreds-method/>
 - OID4VP In-Task Authorization Extension for A2A protocol: <https://github.com/a2aproject/experimental-ext-oid4vp-auth/blob/main/v1/spec.md>

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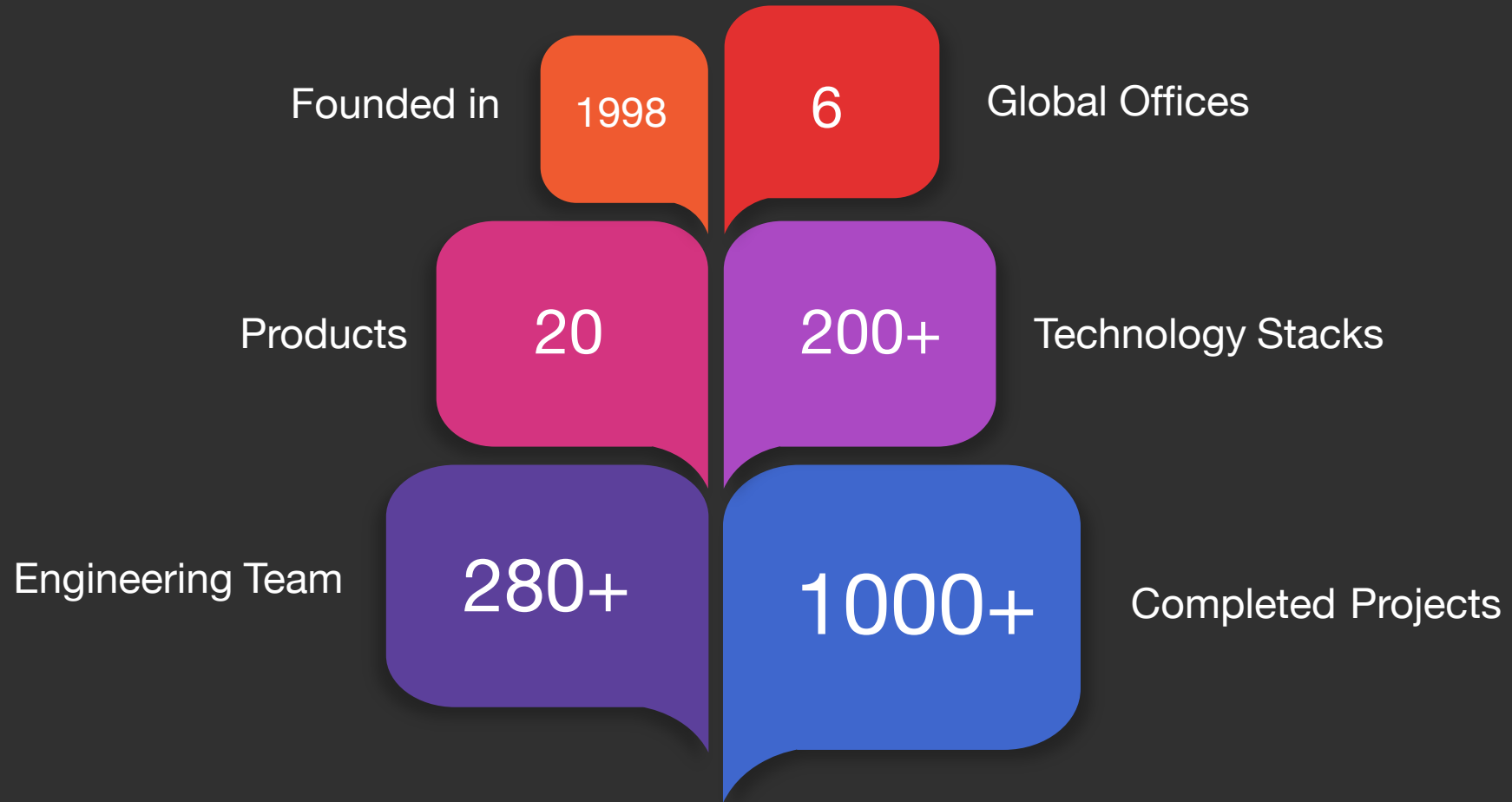




Q&A

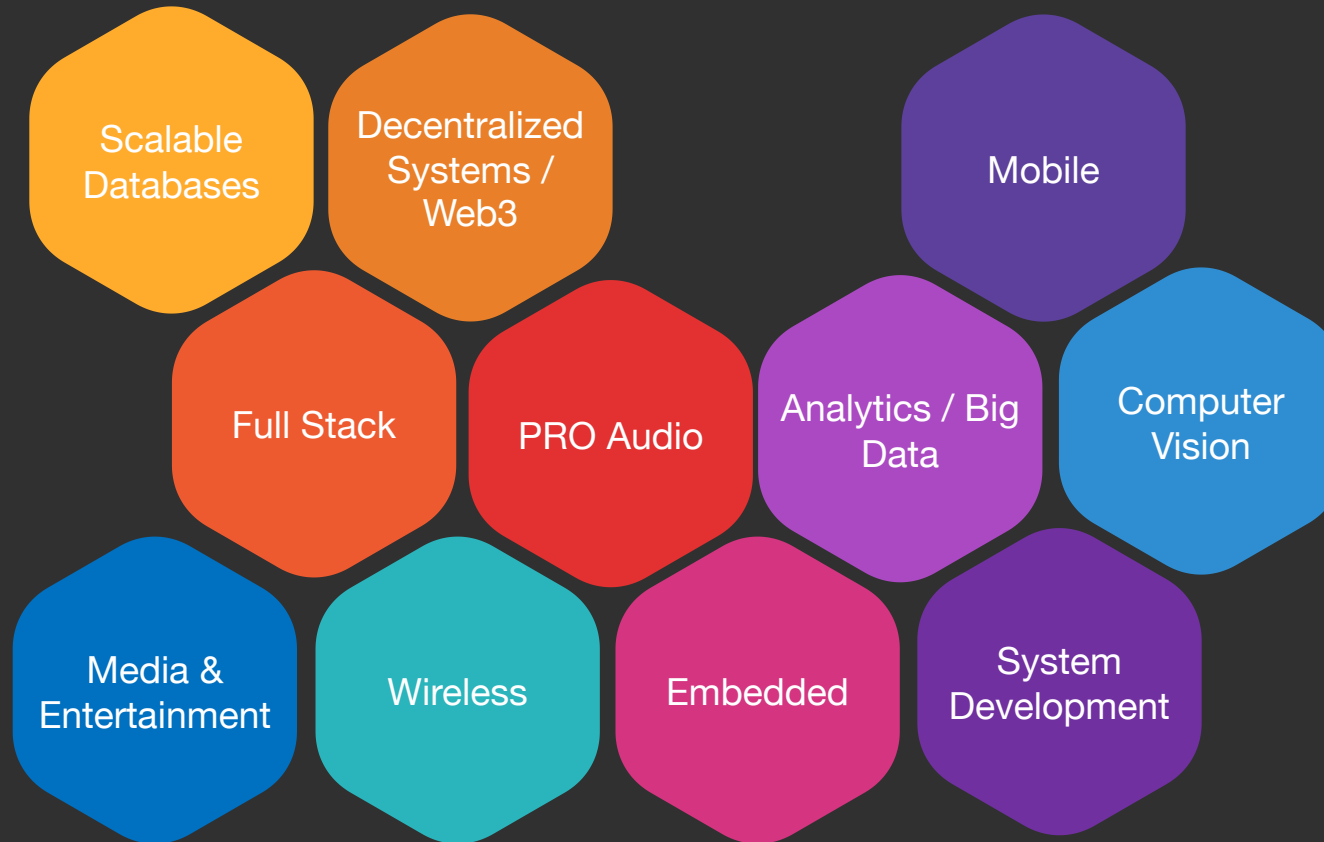


Appendix – About DSR



Privately-owned US company - custom, complex software development done right

DSR Capabilities



Industry Affiliations



Bluetooth SIG



CSA



Thread Group



Open Wallet
Foundation



HYPERLEDGER

Hyperledger
Foundation



iOS/Android
Developer



MFi Licensee



Trust over IP
Foundation



Decentralized
Identity Foundation



The Linux
Foundation

Industries Served



Media and
Entertainment



IoT



Financial and
Business Services



Marketing and
Advertising



Consumer
Electronics



Healthcare /
Medical



Avionics



Automotive



Telecom

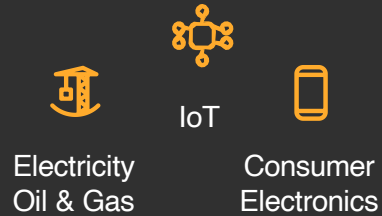


Server &
Data Storage

Industries – Products & Services – Customers



Energy and Consumer Electronics



- Building access control system
- Home security monitoring system (Zigbee)
- Smart metering device
- Participation in development of the Smart Energy Zigbee standard

Transportation



- Toyota Prius 2014 infotainment system
- VW car component
- Aircraft configuration design system
- Wireless communication for highways (802.11p)
- Test suite for ARINC 653, DO-178C certifications

Data Processing



- Secure Document Storage SAAS
- Universal Laboratory Database web-based system
- Server farm infrastructure support system (80,000+ servers)

Web & Telecom Communication



- Dalet system support/maintenance
- Media Asset Management (MAM) web-based system
- Audio quality control system
- IMF format implementation for Netflix

Representative Clients



Quantum.

ORACLE®

CASIO.

SONY

amazon

EMULEX®

NORDIC
SEMICONDUCTOR

Panasonic

SEAGATE

FUJITSU
COMPUTERS
SIEMENS

SHARP.

TOSHIBA

SALTSTACK

Card Access | ENGINEERING

Adaptive
COMPUTING

FORBIDDEN

e•ernym

prodatakey™

SYSGO
EMBEDDING INNOVATIONS

FRONT PORCH DIGITAL

LABWORKS

qorvo™

arm

legrand®

DOING SOFTWARE RIGHT

Full Stack Web

Embedded

Wireless

Analytics / Big Data

Scalable Databases

Digital Media

System Software

Mobile

Blockchain

CV / Machine Learning